

UNIVERSITY OF MORATUWA

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BSc Engineering Honours Degree Semester 4 Examination: 2022

CS3022: SOFTWARE ENGINEERING

- **Time allowed: 2 Hours**
- **ADDITIONAL MATERIAL: None**
- **September 2022**

INSTRUCTIONS TO CANDIDATES:

1. This paper consists of 3 questions in 2 pages.
2. Answer all 3 questions.
3. Start answering each of the 3 main questions on a new page.
4. The maximum attainable mark for each question is given in brackets.
5. This examination accounts for 60% of the module assessment.
6. This is a close-book examination. **NB:** It is an offence to be in possession of unauthorised material during the examination.
7. Only calculators approved and labelled by the Faculty of Engineering are permitted.
8. Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions made on the script.
9. In case of any doubt as to the interpretation of the wording of a question, make suitable assumptions and clearly state them on the script.
10. This paper should be answered only in English.

Question 1 (25 marks)

(i) Using the recently introduced National Fuel Pass system in Sri Lanka as an example, explain the functional, non-functional and domain requirements of a given system. [6 marks]

(ii) Software process models help us to manage the software lifecycle. Critically examine the future of software process models and their importance. [5 marks]

(iii) Describe the following:

- a. User Acceptance Test.
- b. Service Oriented Architecture
- c. Software Evolution
- d. Requirement amalgamation [4 × 2 \text{ marks}]

(iv) Using a travel and tourism booking solution explain how the Web Services can be used with a suitable design diagram to support your answer. [6 marks]

Question 2 (50 marks)

The Ministry of Power and Energy is planning on developing a comprehensive solution for managing the fuel crisis in Sri Lanka. As a part of the solution, they want to address the refined fuel and crude oil procurement from vendors, crude oil distribution to the refineries, fuel distribution to the fuel stations and other large-scale consumers in the country (eg: CEB, SLTB, Exporters, etc.), managing fuel distribution from the fuel stations to the retail customers (general public), and providing limited notifications to the customers regarding fuel availability (notifications are to be generated conservatively to avoid panic and spreading of misinformation). In addition, the solution should also contain a dashboard to view the real-time status of the refined fuel in the country, the status of the crude oil refinement in the country as well as generating predictions and reports for planning and decision making. The retail distribution component of the system is expected to be first deployed within the Colombo municipal council with 500000 users for two weeks using geo restriction. Then the system is expected to be expanded district-wise until the countrywide coverage is achieved. You have been recruited as a lead software engineer for the project by the competent software development company that is developing the solution.

- (i) Identify five key stakeholders of this system. Briefly justify your answer. [5 marks]
- (ii) Draw Five use case diagrams each for the identified stakeholder. Text descriptions are not required. [5 marks]
- (iii) State and describe three (3) non-functional requirements for the system. [3 marks]
- (iv) Draw a suitable UML Activity diagram for the retail customer registration process. State your assumptions. [3 marks]
- (v) Briefly describe the Architecture at large of the proposed system using suitable illustrations. [2 marks]
- (vi) Prepare complete architecture diagrams using the following perspectives for the proposed system with sufficient finer level details.
 - a. Software component level breakdown using a class diagram [5 marks]
 - b. Software component level interactions using sequence diagrams [6 marks]
 - c. High-level structure using a package diagram [4 marks]
- (vii) To improve the rapid development of the solution, you have decided to introduce DevOps practices to the development process.
 - a. Name three (3) DevOps best practices that you can introduce for the development process. [3 marks]
 - b. Identify a CI/CD pipeline for the proposed system and draw the steps of the pipeline as a simple flow chart. [5 marks]
 - c. What are the advantages of using Infrastructure as Code in your development process? [2 marks]
- (viii) Based on the description given and considering the frequent updates that will be introduced by the rapid development process, what is the most suitable software deployment strategy/pattern? Briefly justify your answer based on the requirements of the system. [5 marks]

(ix) Name two key benefits of frequently rewriting large-scale software systems. [2 marks]

Question 3 (25 marks)

(i) Algorithmic costing and Expert estimations are the two main forms of software pricing. One of your project clients demands to use both in a combined manner for the effort estimation of their projects. Using the base models of both approaches, derive a combined estimation model to be used in the given project. Clearly state your assumptions and indicate the work to derive your model. [5 marks]

(ii) As the industry data indicate software engineers often change their employing organizations more frequently than other professions. Analyze this using Maslow's hierarchy of needs. [5 marks]

(iii) Risk Management is an important task for making a software project successful. Considering Risk severity and Risk likelihood as the key parameters prepare a generic risk management model to be used in software projects to manage risks and provide suitable examples wherever relevant. [5 marks]

(iv) Discuss the need for version control. Your answer must explain the codeline, baseline, merging and branching. [5 marks]

(v) It is said, "last two decades we have seen software replacing many systems, jobs, practices, etc.; and in the next two decades we will witness the disappearance of custom software due to AI systems". Critically examine this statement and elaborate on how you would safeguard your profession as a software engineer in such a context. [5 marks]

END OF THE PAPER